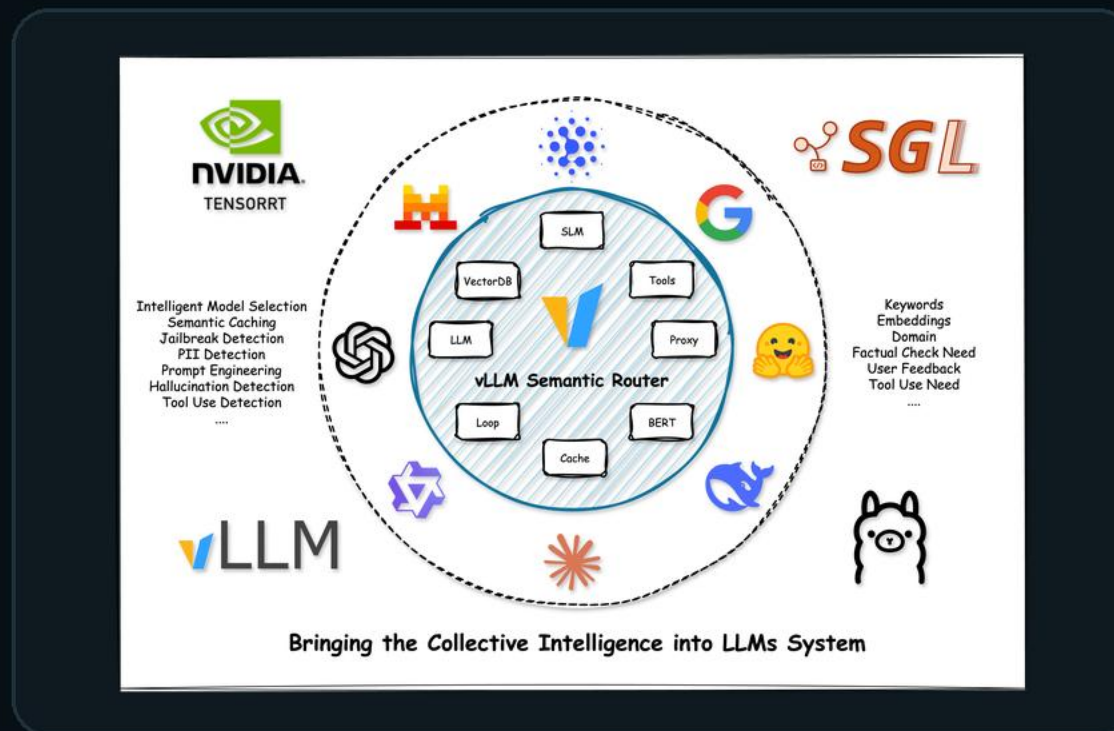


vLLM Semantic Router Initial Project / 初始專案架構提案

- Message: present vLLM Semantic Router as a signal-driven control plane for routing, safety, observability, and edge fleet governance.



Technical Thesis / 技術主張

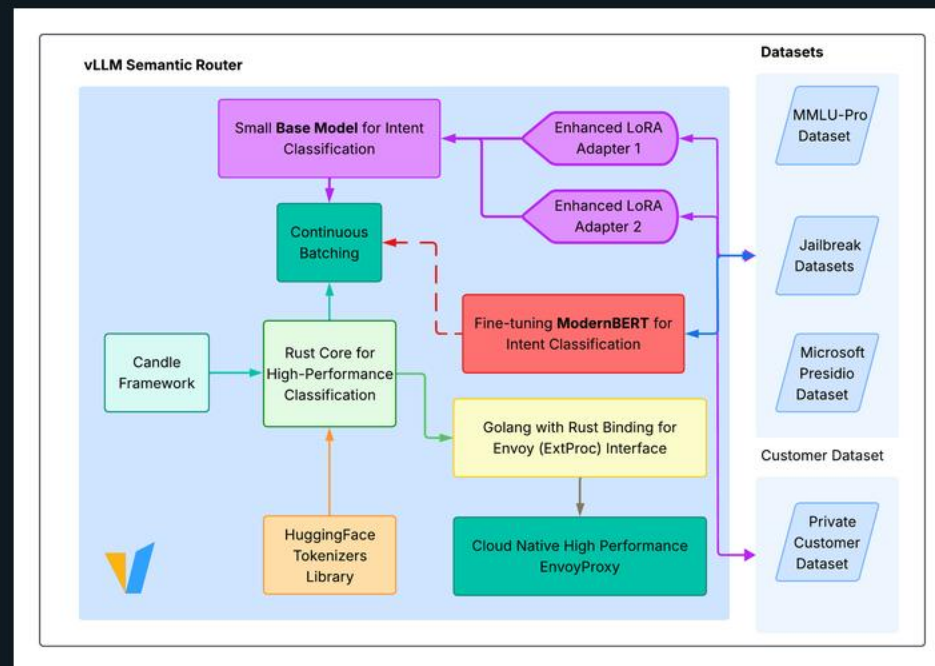
- Message: a router becomes useful when it interprets request signals, composes decisions, and routes to the right execution lane with traceability.
- Customer takeaway: this is not a single model demo; it is an operational layer for multi-model systems.

01 Message: a router becomes useful when it interprets request

02 Customer takeaway: this is not a single model demo; it is an

System Architecture / 系統架構

- Message: requests are decomposed into signals, projected into decisions, optionally transformed by plugins, then routed to model targets.



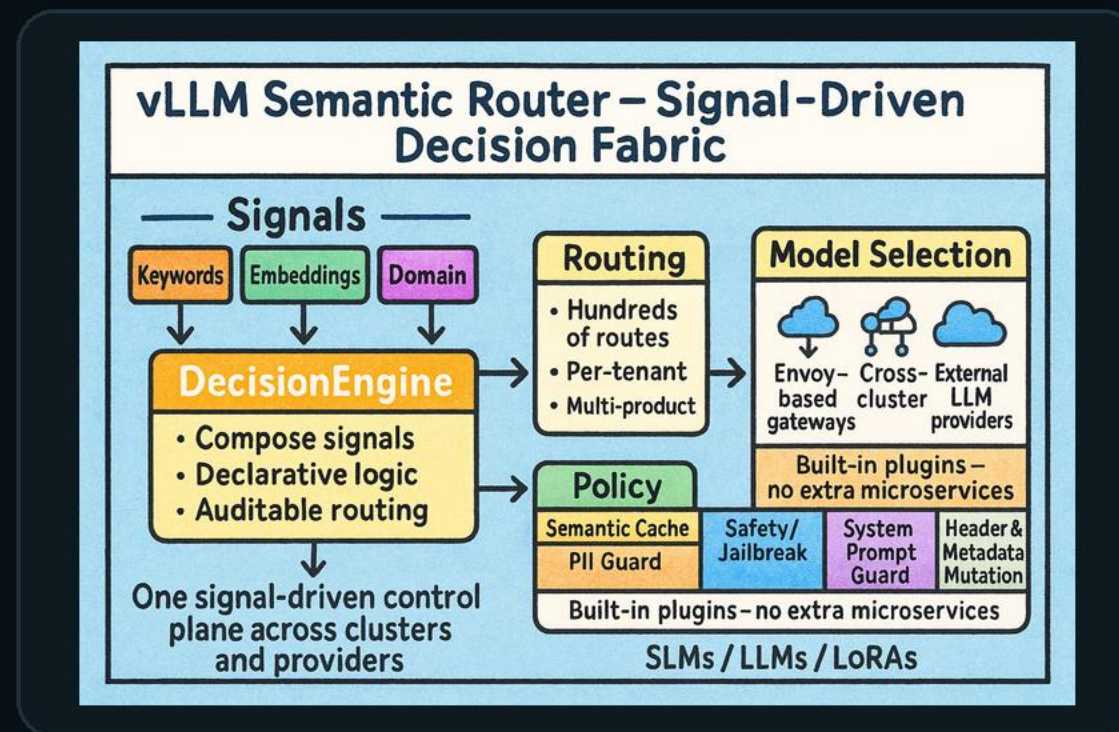
Runtime Data Path / 執行期資料路徑

- Message: Envoy receives OpenAI-compatible traffic, calls the semantic router through ExtProc, and forwards the enriched request to backend providers.

01 Message: Envoy
receives
OpenAI-compatible

Decision Fabric / 決策織物

- Message: declarative decision rules make routing understandable, portable, and auditable across products and deployment environments.



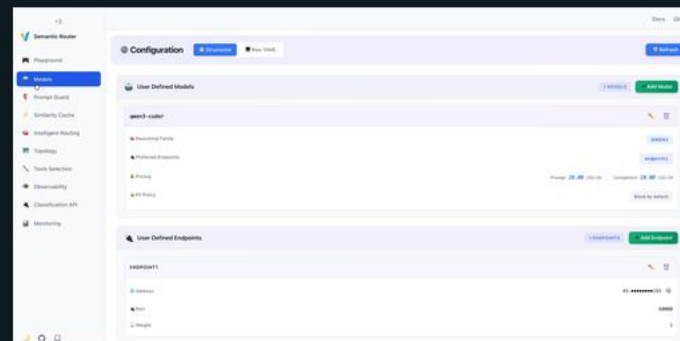
Safety and Governance Lanes / 安全與治理路由

- Message: PII and jailbreak detections are signals; policy decisions can trigger fast responses, response guards, or protected model lanes.

01 Message: PII and jailbreak detections are signals; policy decisions

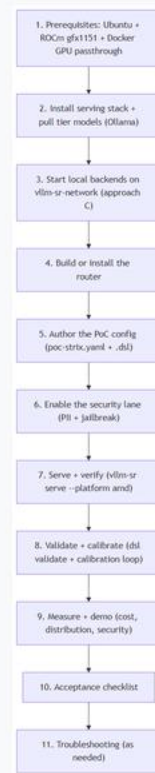
Observability and Explainability / 可觀測性與可解釋性

- Message: every route can expose diagnostic headers and dashboard views so customers can inspect model selection, signals, and savings.



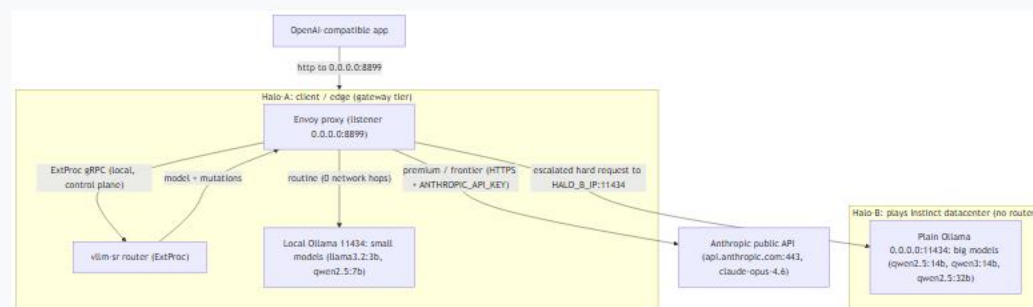
Strix Halo Single-Box PoC / 單機 PoC

- Message: one Strix Halo box runs the full local stack: Ollama ROCm, five tier models, router, safety lane, smoke tests, and offline validation.



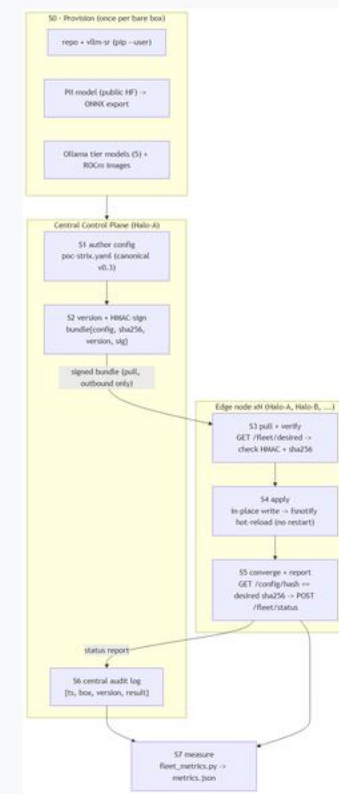
2-Box Edge Gateway / 雙機邊緣閘道路由

- Message: keep routine traffic local at the edge and escalate hard tiers to a second box or external provider without double-hop routing.



Fleet Control Plane / 艦隊控制平面

- Message: a pull-mode central control plane distributes signed config to bare edge gateways with central audit and drift recovery.



Convergence Contract / 收斂契約

- Message: the Python CCP signs sha256(config_bytes); the Go router reports /config/hash; agents apply config in place so fsnotify hot-reloads without restart.

01 Message: the Python
CCP signs
sha256(config_bytes);

Verified Evidence / 已驗證成果

- Message: the fleet design is hardware-verified on two real Strix Halo gateways.

Verified figures:

- Run bundle: run-20260701-154843
- Both real routers converged to signed hash a78aebc5fd5f
- One central edit converged both routers to fc739baa...
- Rollback returned both routers to a78aebc5fd5f
- Five converged versions: v1 to v5
- Cross-box convergence span: 0-3s, bounded by poll interval
- Halo-A router cold-start: 565s

01 Message: the fleet design is hardware-verified on

02 Run bundle: run-20260701-154843

03 Both real routers converged to signed hash a78aebc5fd5f

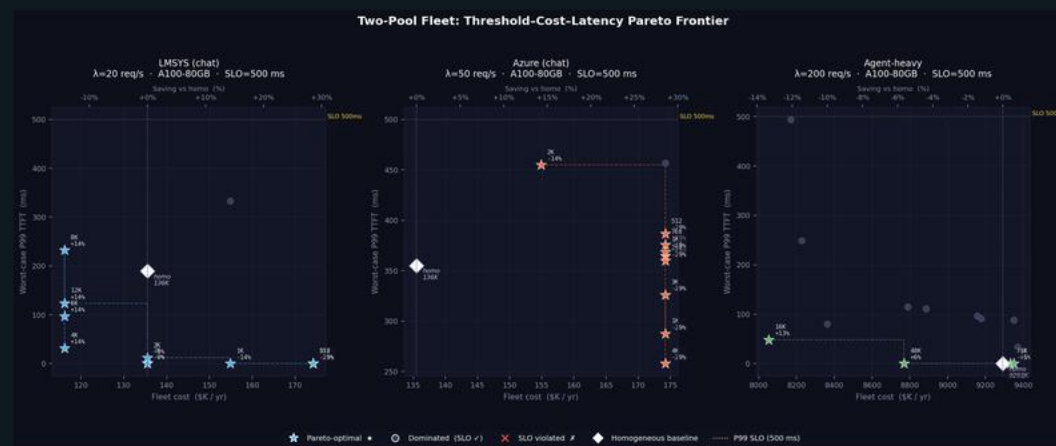
04 One central edit converged both routers to fc739baa...

05 Rollback returned both routers to a78aebc5fd5f

06 Five converged versions: v1 to v5

Benchmark and TCO Context / Benchmark 與 TCO 脈絡

- Message: the benchmark stack and fleet-sim provide the path from replay evidence to TCO analysis, while hardware-specific performance claims remain explicitly scoped.
- Caveat: fleet-sim TCO is simulation/extrapolation unless rerun on target hardware.

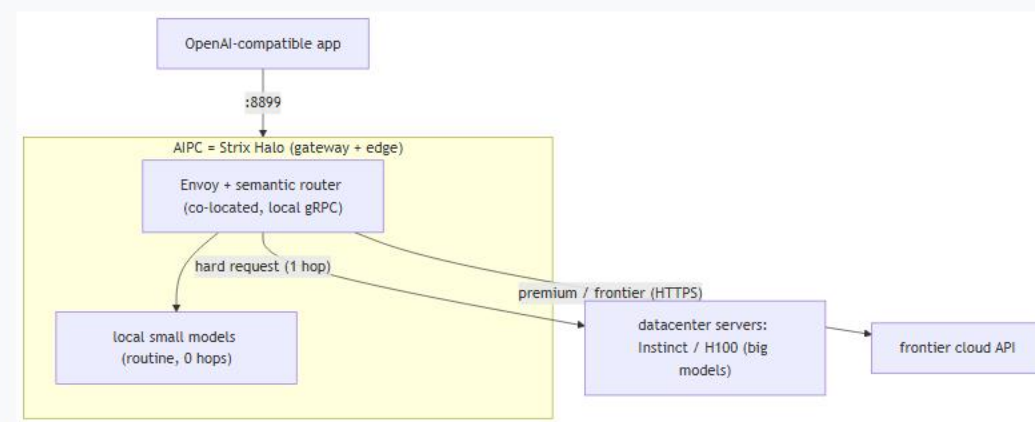


Initial Project Scope / 初始專案範圍

- Message: propose a controlled customer pilot that starts with one routing profile, one safety lane, one observability loop, and one deployment topology.

Suggested deliverables:

- customer traffic taxonomy and signal map
- routing config and DSL review
- single-box or 2-box deployment rehearsal
- dashboard and replay evidence
- final architecture and operating runbook



Risks and Mitigations / 風險與緩解

- Message: the key risks are operational, not architectural: image/version skew, model staging, network reachability, and finer-grained hot-reload metrics.

Mitigation anchors:

- pin router images or migrate config schema deliberately
- preflight model assets and PII ONNX export
- verify container-to-host reachability for remote tiers
- add write-to-converge timers for sub-second hot-reload measurement

01 Message: the key risks are operational, not architectural:

02 pin router images or migrate config schema deliberately

03 preflight model assets and PII ONNX export

04 verify container-to-host reachability for remote tiers

05 add write-to-converge timers for sub-second hot-reload

Roadmap / 下一步

- Message: scale the customer pilot into a governed multi-model control plane with better measurement, broader fleet patterns, and customer dataset validation.

Roadmap items:

- sub-second reload/convergence instrumentation
- customer replay and data-quality calibration
- broader fleet and operator alignment
- cost/latency policy tuning
- production readiness review

01 Message: scale the customer pilot into a governed multi-model

02 sub-second reload/convergence instrumentation

03 customer replay and data-quality calibration

04 broader fleet and operator alignment

05 cost/latency policy tuning

06 production readiness review